PBQ Infection Analysis — Study Handout

Teacher-style answer key with concise reasoning

Origin of Infection	**PC-4**	**Malicious IP**	**198.51.100.77**

Device Classification

Host	Status	Why
PC-1	Clean	Normal user activity only; no suspicious process or bad IP traffic.
PC-2	Infected	Inbound 445 allowed, security service stopped, suspicious process ran, then bad IP beacon.
PC-3	Clean	Normal browsing and file activity only; no lateral movement or beaconing.
PC-4	Infected	Downloaded ZIP, extracted EXE, executed it, then connected to the bad IP and scanned internally.
PC-5	Clean	Targeted over 445, but payload was quarantined and execution was blocked.
PC-6	Clean	445 attempt was blocked at the firewall; no successful compromise.
PC-7	Clean	Normal intranet/web access only; no malicious indicators.
PC-8	Infected	445 allowed, admin account created, added to Administrators, then bad IP beacon.

Core Logic to Remember

• PC-4 is **patient zero**: download → extract → execute → outbound connection.

• **198.51.100.77** is the malicious IP because multiple infected hosts beacon to it.

• Internal spread uses **TCP 445 / SMB**, a classic lateral movement path.

• Being **targeted** does not automatically mean infected: PC-5 stayed clean because the payload was quarantined; PC-6 stayed clean because the connection was blocked.

• The truly infected hosts are the ones with **successful execution, persistence/admin changes, disabled defenses, or outbound C2**.

Ultra-Short Memorization

Origin: PC-4 Bad IP: 198.51.100.77 Infected: PC-2, PC-4, PC-8 Spread: SMB / 445